PC-202: Theory of Computation

Unit-I

Automata and Language Theory: Chomsky Classification, Finite Automata, Deterministic Finite Automata (DFA),Non-Deterministic Finite Automata (NFA), Regular Expressions, Equivalence of DFAs, NFAs and Regular Expressions, Algebraic laws for Regular expressions, Kleen's Theorem, Regular expression to FA, DFA to Regular expression,Arden Theorem, Closure properties of Regular grammar, Non-Regular Languages, Pumping Lemma, Minimization of Finite Automata, Distinguishing one string from other, Myhill-Nerode Theorem, Decision properties of Regular Languages, FA with output: Moore and Mealy machine, Equivalence of Moore and Mealy Machine, Applications and Limitation of FA.

Unit-II

Context Free Languages: Context Free Grammar (CFG), Parse Trees, Ambiguity in Grammar, Ambiguous to Unambiguous CFG, Parsing, LL(K) grammar, Simplification of CFGs: Elimination of Useless symbols, epsilon productions, and unit productions from CFGs, Normal forms for CFGs: CNF and Greibach Normal Form (GNF), Push Down Automata (deterministic and non-deterministic) (PDA), Equivalence of CFGs and PDAs, CFG to PDA and PDA to CFG, two stack PDA, Pumping Lemma, Closure properties of CFLs, decision problems for CFLs.

Unit-III

Turing Machines and Computability Theory: Definition, design and extensions of Turing Machine, Equivalence of various Turing Machine Formalisms, Church – Turing Thesis, Decidability, Halting Problem, Reducibility and its use in proving undecidability, Rices theorem, Undecidability of Posts correspondence problem., Recursion Theorem and its significance, Classification of languages: Recursive and Recursively Enumerable (r.e.) languages, existence of non-recursively enumerable (non-r.e.) languages, informal proofs demonstrating unsolvability of certain computational problems.

Unit-IV

Computational Complexity Classes: Definition of class P as the consensus class of tractable problems, Classes NP, co-NP, and their significance, Polynomial-time reductions and their role in complexity theory, NP-completeness and NP-hardness with examples, Cook-Levin Theorem (including proof) and its implications, NP-complete problems in different domains: Graph problems: Clique, Vertex Cover, Independent Set, Hamiltonian Cycle, number problem (partition), set cover, Space Complexity: PSPACE and NPSPACE complexity classes, Savitch's Theorem (with proof) and its impact on space complexity, Probabilistic Computation and the BPP Class, Interactive Proof Systems and the IP Class, Relativized Computation and the concept of oracles.

PC-204: Software Engineering

UNIT - I

Introduction: Software Components, Software Characteristics, Software Crisis, Software processes and its models (waterfall, incremental development, spiral model, re-use-oriented model, prototype), An Introduction to Non-Traditional Software Development Process: Rational Unified Process, Rapid Application Development. Process activities, Process improvement (CMM Levels), Software Reuse. Agile Development model, plan driven vs agile model of development, agile methods and development techniques (user stories, refactoring, test first development, pair programming, agile project management (SCRUM agile method).

UNIT - II

Requirement Engineering: Functional and non-functional requirements, requirement elicitation, use case development, requirement analysis and validation, requirement review or requirement change, SRS document. Size Estimation: Software Size, LOC and function point, cost and effort estimation, COCOMO, ISO 9001:2015 Certification, Halstead's Metrics & Software Measurement Techniques, Reliability Metrics & Models, Risk Management & Scheduling Techniques (PERT, Gantt Charts), CASE Tools for Requirement Engineering. System modelling: Interaction models: Use case diagram, sequence diagrams, Structural models: class diagrams, generalization, aggregation, Behavioural models: ER diagrams, Data flow diagrams, data dictionaries.

UNIT - III

Software Design: Architectural views and patterns, Modularity (cohesion and coupling), information hiding, functional independence, function-oriented design, object-oriented design, SOA, SAAS, Design Patterns (Singleton, Factory, Observer), UML Modelling (Class, Sequence, State Diagrams). Software Quality: McCall's Quality Factors, ISO 9126 Quality Factors, Quality Control, Quality Assurance, Software Reliability, Software Configuration Management. Software Evolution: Evolution process, legacy system, Software maintenance: Maintenance prediction, Software Maintenance Strategies (Corrective, Adaptive, Perfective, Preventive Maintenance), Re-Engineering, Reverse Engineering, Refactoring.

UNIT –IV

Software Testing: verification, validation, Development testing (unit testing, component testing, system testing,Test Driven Development (TDD), Release Testing (Requirement based testing, scenario testing, performance testing), User testing (alpha, beta and acceptance testing), Regression Testing, Stress Testing, Mutation Testing & Fuzz Testing. System Security: Reliability engineering, reliability requirements (functional and non-functional) and its measurement, safety engineering: safety critical systems, its requirement, security engineering and its requirements, security guidelines

PC 206 :Computer Networks

UNIT I

Overview, Types of computer networks: LAN, MAN, WAN, Wireless and Wired networks, broadcast and point-to-point networks, Network topologies; protocol suites: TCP/IP and OSI, History, Standard.Physical Layer and Transmission Media: Data and Signals, Digital Transmission, Analog Transmission, Bandwidth utilization, Multiplexing : Frequency Division, Time Division, Wavelength Division, Transmission Media, Switching:Circuit Switching, Message Switching, Packet Switching.

UNIT II

Data-Link Layer (Wired Networks): Introduction, DLC, Multiple Access Protocols, Wired LANS (Ethernet, others)Data-Link Layer (Wireless Networks): Introduction, IEEE 802.11, Bluetooth, WiMAX, Cellular telephony, Satellite Networks, Mobile IP.Transport Layer: Protocols: simple, stop-and-wait, GBN, Selective repeat, Bidirectional protocols, Internet Transport Layer protocols, UDP, TCP

UNIT III

Network Layer: Introduction, Addressing : Internet address, subnetting, IPv4, ICMPv4, Unicast Routing, Multicast routing, IPV6, ICMPv6.Application Layer: Application layer paradigm, Client-server paradigm, Standard Client Server Applications, P2P,Socket Interface programming.

UNIT IV

Multimedia and QoS: Data types, streaming of audio/video, real-time interactive protocols, Quality of Service.Network Management: Introduction, SNMP, ASN.1 Security: Introduction, Ciphers, Application layer security, transport layer security, network layer security, packet filter firewall, proxy firewall. Programming: Socket programming.

PC 208:Artificial Intelligence & Machine Learning

Unit-1

Introduction to Artificial Intelligence: Definition and goals of Artificial Intelligence,

History and evolution of AI,Intelligent agents and environments, Problem-solving using AI techniques, Turing test and its significance. Problem Solving and Search Algorithms: Problem formulation and state space representation; Uninformed search algorithms: Breadth-First Search, Depth-First Search, Uniform-Cost Search; Informed search algorithms: A* Search,Best-First Search; Heuristic functions and their properties; Constraint satisfaction problems and backtracking algorithms

Unit-2

Introduction to Machine Learning: Paradigms of Machine Learning- Supervised learning, Unsupervised learning,Semi-supervised learning, Active learning, Self-supervised learning, Transfer learning, Domain adaptation, Zero-shot, One-shot and Few-shot learning; Federated learning.

Unit-3

Supervised learning: Decision trees, linear classifiers and kernels, linear regression; Naïve Bayes, k-Nearest Neighbors; Unsupervised learning: Clustering and dimensionality reduction; Expectation Maximization,Dimensionality Reduction. Introduction to neural networks and deep learning; Feature extraction and selection, PCA, factor analysis, manifold learning; Evaluation metrics for machine learning models.

Unit-4

Knowledge Representation and Reasoning: Propositional and first-order logic; Knowledge representation using semantic networks, frames, and ontologies; Resolution and inference in propositional logic; Bayesian networks and probabilistic reasoning; Common-sense reasoning and expert systems. Applications &Research Topics: Application in the field of web and data mining, text recognition, speech recognition.

PC 210: Web Technologies

Unit -1

HTML and CSS: HTML Vs HTML5, Basics of HTML, HTML Document Structure. Understand indentation and nesting in HTML code. Learn to use HTML tags to structure headings, paragraphs. Creating unordered and ordered lists.HTML Tables, Tables Attributes, Rows and Columns. Inserting Images, creating hyperlinks using anchor tags. Introduction to Forms, The <FORM>, <INPUT> Tag Text input. Create multi-page websites. To Learn HTML best practices. Introduction to cascading style sheets and its use, CSS selectors and properties. Types of CSS: inline, internal and external CSS. CSS specificity and inheritance, the CSS Box Model, CSS positioning and display properties.Font styling, CSS float and clear properties.

Unit-2

JavaScript: Overview and its uses, Basic syntax and data types, Operators and expressions, including arithmetic, comparison, and logical operators, Control structures like if/else statements and loops (The for loop and while loop),Function declaration and expression, Higher-order functions, including: Passing functions as arguments to other functions, Returning functions from other functions. The map, filter, and reduce methods on arrays, Creating and accessing arrays, Adding and removing elements from arrays, The slice method for slicing arrays, the concat method for concatenating arrays, The indexOf and lastIndexOf methods for finding elements inarrays.Object-oriented programming in JavaScript: Creating objects with object literals, Creating objects with constructor functions and the new keyword, Creating objects with classes and the class keyword, Adding and accessing properties and methods on objects, The this keyword and how it works in object methods, The super keyword for accessing parent class methods and properties. Manipulating objects and arrays using methods and iteration, including: The Object. Keys and Object.values methods.

Unit-3:React.JS: front-end development with React, Understand when and how to use React Components, Learn to pass Props and work with them, write JSX and understand JSX syntax, the React DOM, State Management in React, React Hooks, conditional rendering in React, Understand the difference between class and functional components. NODE.JS: the components of back-end development, workingwith an MVC framework, Apply concepts like data types, objects, methods, object-oriented programming, and classes in the context of back-end development. Server-Side JavaScript, Using Node on the command line, NPM, JavaScript Build Processes, Event Loop and Emitters File System Interaction, Modules, Native Node drivers.

Unit-4: DataBase: Working and connecting with Database, Persistent connection use RDBMS. Authentication and Security: the need for authentication and keeping userdetails secure, Encryption and use encryption to keep your database secure, implement Hashing and Salting with bcrypt, Using Sessions and Cookies to persist user log in sessions. Setting up local authentication from scratch. Implementing Passport to authenticate users quickly and effectively. Understand and use environment variables to keep secret keys secure, use OAuth 2.0 to log in users using Google.

AEC 212: Engineering Economics

UNIT I

Introduction, Flow in an economy, Law of Supply and Demand, Concept of Engineering Economics, Elements of Cost, Break-Even Analysis, P/V ratio, examples of simple economic analysis, Interest Formulas and Their Applications.

UNIT II

Present Worth Method of Comparison: Introduction, Revenue Dominated Cash Flow Diagram, Cost- Dominated Cash Flow Diagram Future Worth Method: Introduction, Revenue Dominated Cash Flow Diagram, Cost-Dominated Cash Flow Diagram Annual Equivalent Method: Introduction, Revenue Dominated Cash Flow Diagram, Cost-Dominated Cash Flow Diagram, Alternate approach. Rate of Return Method.

UNIT III

Replacement and Maintenance Analysis: Introduction, Types, Determination of economic life of an asset ,replacement method. Depreciation: Introduction and methods of depreciation (Straight line, Declining Balance, Sum of the Years Digitmethod, Sinking fund method, Service output method). Evaluation of public alternative.

UNIT IV

Inflation Adjustment: Introduction, Procedure to adjust Inflation, Inflation Adjusted Economic Life of Machines. Inventory Control and Methods, Make or buy decision, Project Management: Introduction, Phases, CPM, Gantt/Time Chart, PERT. Value Analysis / Value Engineering